

**INDIAN INSTITUTE OF TECHNOLOGY (INDIAN SCHOOL OF MINES)
DHANBAD**

OFFICE OF THE DEAN (ACADEMIC)

No. IITISM/DAC/ 1103

18th Mar, 2026

NOTIFICATION

**LIST OF COURSES REQUIRED FOR THE COMPLETION OF RESPECTIVE
MINOR/HONORS OFFERED BY THE VARIOUS DEPARTMENTS UNDER NEP
REGIME.**

The Senate in its 40th meeting approved the updated list of courses for various Minor / Honor being offered under NEP regime.

The relevant documents are attached herewith for reference.


18/03/2026
Dean (Academic)

Copy to:

1. Director/ Dy. Director
2. Registrar
3. All Deans/ Associate Deans/ PIC(Automation)
4. All HODs
5. JR (Acad)/ DR(UG)/ AR(PG)

SL	dept_id	course_title	minor/honor	sub_code	sub_name	lecture	tutorial	practical	credits	contact_hours	sub_type	Course Type
1	agl	Geology	Minor	NGLC201	Crystallography and Optical Mineralogy	3	1	0	4	4	Theory	DC (Minor)
2	agl	Geology	Minor	NGLM301	Crystallography and Optical Mineralogy Practical	0	0	3	1.5	3	Practical	SDC (Minor)
3	agl	Geology	Minor	NGLC205	Petrogenesis of Igneous Rocks	3	1	0	4	4	Theory	DC (Minor)
4	agl	Geology	Minor	NGLM302	Igneous Petrogenesis Practical	0	0	3	1.5	3	Practical	SDC (Minor)
5	agl	Geology	Minor	NGLC503	Methods of Structural Geology	3	1	0	4	4	Theory	DC (Minor)
6	agl	Geology	Minor	NGLM401	Introduction to Structural Geology Practical	0	0	3	1.5	3	Practical	SDC (Minor)
7	agl	Geology	Minor	NGLC206	Metamorphic Petrology	3	1	0	4	4	Theory	DC (Minor)
8	agl	Geology	Minor	NGLC207	Remote Sensing and GIS	3	1	0	4	4	Theory	DC (Minor)
9	agl	Geology	Minor	NGLM402	Basics of Metamorphic Petrology Practical	0	0	3	1.5	3	Practical	SDC (Minor)
10	agp	Geophysical Technology	Minor	NGPC503	Computational Seismology	3	1	0	4	4	Theory	DC (Minor)
11	agp	Geophysical Technology	Minor	NGPC506	Computational Seismology Practical	0	0	3	1.5	3	Practical	SDC (Minor)
12	agp	Geophysical Technology	Minor	NGPC507	Remote Sensing: Principles and Data Acquisition System	3	1	0	4	4	Theory	DC (Minor)
13	agp	Geophysical Technology	Minor	NGPC510	Remote Sensing: Principles and Data Acquisition System Practical	0	0	3	1.5	3	Practical	SDC (Minor)
14	agp	Geophysical Technology	Minor	NGPC502	Geotechnical Modelling	3	1	0	4	4	Theory	DC (Minor)
15	agp	Geophysical Technology	Minor	NGPC504	Geotechnical Modelling Practical	0	0	3	1.5	3	Practical	SDC (Minor)
16	agp	Geophysical Technology	Minor	NGPC508	Advanced Numerical Methods	3	1	0	4	4	Theory	DC (Minor)
17	agp	Geophysical Technology	Minor	NGPC509	Earthquake Statistics and Hazard	3	1	0	4	4	Theory	DC (Minor)
18	agp	Geophysical Technology	Minor	NGPC511	Advanced Numerical Methods Practical	0	0	3	1.5	3	Practical	SDC (Minor)
19	ce	Chemical Engineering	Honor	NCHH301	Optimization in Chemical Engineering	3	1	0	4	4	Theory	DC (Hons)
20	ce	Chemical Engineering	Honor	NCHH302	Computational Methods Laboratory	0	0	3	1.5	3	Practical	DP (Hons)
21	ce	Chemical Engineering	Honor	NCHH303	Computational Fluid Dynamics	3	1	0	4	4	Theory	DC (Hons)
22	ce	Chemical Engineering	Honor	NCHH304	Optimization and AI in Chemical Engineering Laboratory	0	0	3	1.5	3	Practical	DP (Hons)
23	ce	Chemical Engineering	Honor	NCHH401	Petroleum Refining	3	1	0	4	4	Theory	DC (Hons)
24	ce	Chemical Engineering	Honor	NCHH402	Process Simulation Laboratory	0	0	3	1.5	3	Practical	DP (Hons)
25	ce	Chemical Engineering	Honor	NCHH403	Heterogeneous Reaction Engineering	3	1	0	4	4	Theory	DC (Hons)
26	ce	Chemical Engineering	Honor	NCHH404	Industrial Safety and Hazards Management	3	1	0	4	4	Theory	DC (Hons)
27	ce	Chemical Engineering	Honor	NCHH405	Fuel Analysis Laboratory	0	0	3	1.5	3	Practical	DP (Hons)
28	ce	Chemical Engineering	Minor	NCSM301	Fundamentals of Chemical Engineering	3	1	0	4	4	Theory	DC (Minor)
29	ce	Chemical Engineering	Minor	NCSM302	Unit Operations Laboratory - I	0	0	3	1.5	3	Practical	SDC (Minor)
30	ce	Chemical Engineering	Minor	NCSM303	Chemical Process Principles	3	1	0	4	4	Theory	DC (Minor)
31	ce	Chemical Engineering	Minor	NCSM304	Unit Operations Laboratory - II	0	0	3	1.5	3	Practical	SDC (Minor)
32	ce	Chemical Engineering	Minor	NCSM401	Chemical Reaction Engineering	3	1	0	4	4	Theory	DC (Minor)
33	ce	Chemical Engineering	Minor	NCSM402	Chemical Reaction Engineering Laboratory	0	0	3	1.5	3	Practical	SDC (Minor)
34	ce	Chemical Engineering	Minor	NCSM403	Fundamentals of Transport Phenomena	3	1	0	4	4	Theory	DC (Minor)
35	ce	Chemical Engineering	Minor	NCSM404	Separation Process Principles	3	1	0	4	4	Theory	DC (Minor)
36	ce	Chemical Engineering	Minor	NCSM405	Process Design Laboratory	0	0	3	1.5	3	Practical	SDC (Minor)
37	cse	Computer Science and Engineering	Honor	NCSH301	Digital Image Processing	3	1	0	4	4	Theory	DC (Hons)
38	cse	Computer Science and Engineering	Honor	NCSH302	Digital Image Processing Laboratory	0	0	3	1.5	3	Practical	DP (Hons)
39	cse	Computer Science and Engineering	Honor	NCSH303	Introduction to Machine Learning	3	1	0	4	4	Theory	DC (Hons)
40	cse	Computer Science and Engineering	Honor	NCSH304	Introduction to Machine Learning Laboratory	0	0	3	1.5	3	Practical	DP (Hons)
41	cse	Computer Science and Engineering	Honor	NCSH401	Number Theory and Cryptography	3	1	0	4	4	Theory	DC (Hons)
42	cse	Computer Science and Engineering	Honor	NCSH402	Number Theory and Cryptography Laboratory	0	0	3	1.5	3	Practical	DP (Hons)
43	cse	Computer Science and Engineering	Honor	NCSH403	Graph Theory	3	1	0	4	4	Theory	DC (Hons)
44	cse	Computer Science and Engineering	Honor	NCSH404	Randomized Algorithms	3	1	0	4	4	Theory	DC (Hons)
45	cse	Computer Science and Engineering	Honor	NCSH405	Randomized Algorithms Laboratory	0	0	3	1.5	3	Practical	DP (Hons)
46	cse	Computer Science and Engineering	Minor	NCSC202	Computer Organization and Architecture	3	1	0	4	4	Theory	DC (Minor)
47	cse	Computer Science and Engineering	Minor	NCSC308	Web Development Laboratory	0	0	3	1.5	3	Practical	SDC (Minor)
48	cse	Computer Science and Engineering	Minor	NCSC305	Computer Networks	3	1	0	4	4	Theory	DC (Minor)
49	cse	Computer Science and Engineering	Minor	NCSC309	Data Presentation and Visualisation Techniques Laboratory	0	0	3	1.5	3	Practical	SDC (Minor)
50	cse	Computer Science and Engineering	Minor	NCSC203	Algorithm Design & Analysis	3	1	0	4	4	Theory	DC (Minor)
51	cse	Computer Science and Engineering	Minor	NCSC403	Computational Techniques Laboratory	0	0	3	1.5	3	Practical	SDC (Minor)
52	cse	Computer Science and Engineering	Minor	NCSC206	Database Management Systems	3	1	0	4	4	Theory	DC (Minor)
53	cse	Computer Science and Engineering	Minor	NCSC208	Operating Systems	3	1	0	4	4	Theory	DC (Minor)
54	cse	Computer Science and Engineering	Minor	NCSC404	Soft Computing Laboratory	0	0	3	1.5	3	Practical	SDC (Minor)

SL	dept_id	course_title	minor/honor	sub_code	sub_name	lecture	tutorial	practical	credits	contact_hours	sub_type	Course Type
55	cve	Civil Engineering	Honor	NCEH301	Advanced Geotechnical Engineering	3	1	0	4	4	Theory	DC (Hons)
56	cve	Civil Engineering	Honor	NCEH302	Advanced Geotechnical Engineering Laboratory	0	0	3	1.5	3	Practical	DP (Hons)
57	cve	Civil Engineering	Honor	NCEH303	Advanced Transportation Engineering	3	1	0	4	4	Theory	DC (Hons)
58	cve	Civil Engineering	Honor	NCEH304	Advanced Transportation Engineering Laboratory	0	0	3	1.5	3	Practical	DP (Hons)
59	cve	Civil Engineering	Honor	NCEH401	Advanced Structural Analysis	3	1	0	4	4	Theory	DC (Hons)
60	cve	Civil Engineering	Honor	NCEH402	Advanced Structural Analysis Laboratory	0	0	3	1.5	3	Practical	DP (Hons)
61	cve	Civil Engineering	Honor	NCEH403	Advanced Design of Structures	3	1	0	4	4	Theory	DC (Hons)
62	cve	Civil Engineering	Honor	NCEH404	Advanced Water Resources Engineering	3	1	0	4	4	Theory	DC (Hons)
63	cve	Civil Engineering	Honor	NCEH405	Advanced Water Resources Engineering Laboratory	0	0	3	1.5	3	Practical	DP (Hons)
64	cve	Civil Engineering	Minor	NCEC201	Geotechnical Engineering	3	1	0	4	4	Theory	DC (Minor)
65	cve	Civil Engineering	Minor	NCEM301	Basic Geotechnical Engineering Laboratory	0	0	3	1.5	3	Practical	SDC (Minor)
66	cve	Civil Engineering	Minor	NCEC208	Structural Analysis -I	3	1	0	4	4	Theory	DC (Minor)
67	cve	Civil Engineering	Minor	NCEM302	Basic Structural Engineering Laboratory	0	0	3	1.5	3	Practical	SDC (Minor)
68	cve	Civil Engineering	Minor	NCEC202	Transportation Engineering	3	1	0	4	4	Theory	DC (Minor)
69	cve	Civil Engineering	Minor	NCEM401	Basic Transportation Engineering Laboratory	0	0	3	1.5	3	Practical	SDC (Minor)
70	cve	Civil Engineering	Minor	NCEC305	Design of Steel Structures	3	1	0	4	4	Theory	DC (Minor)
71	cve	Civil Engineering	Minor	NCEC207	Water Resources Engineering -I	3	1	0	4	4	Theory	DC (Minor)
72	cve	Civil Engineering	Minor	NCEM402	Basic Water Resources Engineering Laboratory	0	0	3	1.5	3	Practical	SDC (Minor)
73	ece	Electronics and Communication Engineering	Honor	NECH301	Optical Engineering	3	1	0	4	4	Theory	DC (Hons)
74	ece	Electronics and Communication Engineering	Honor	NECH302	Optical Engineering Lab	0	0	3	1.5	3	Practical	DP (Hons)
75	ece	Electronics and Communication Engineering	Honor	NECH303	VLSI Technology and Design	3	1	0	4	4	Theory	DC (Hons)
76	ece	Electronics and Communication Engineering	Honor	NECH304	VLSI Design Lab	0	0	3	1.5	3	Practical	DP (Hons)
77	ece	Electronics and Communication Engineering	Honor	NECH401	Industrial Instrumentation	3	1	0	4	4	Theory	DC (Hons)
78	ece	Electronics and Communication Engineering	Honor	NECH402	Industrial Instrumentation Lab	0	0	3	1.5	3	Practical	DP (Hons)
79	ece	Electronics and Communication Engineering	Honor	NECH403	Machine Learning and AI	3	1	0	4	4	Theory	DC (Hons)
80	ece	Electronics and Communication Engineering	Honor	NECH404	Wireless Communication	3	1	0	4	4	Theory	DC (Hons)
81	ece	Electronics and Communication Engineering	Honor	NECH405	Wireless Communication Lab	0	0	3	1.5	3	Practical	DP (Hons)
82	ece	Drone Electronics	Minor	NECC203	Microprocessors & Microcontrollers	3	1	0	4	4	Theory	DC (Minor)
83	ece	Drone Electronics	Minor	NECM301	Microprocessor Lab	0	0	3	1.5	3	Practical	SDC (Minor)
84	ece	Drone Electronics	Minor	NECC208	Principles of Communication Systems	3	1	0	4	4	Theory	DC (Minor)
85	ece	Drone Electronics	Minor	NECM302	Communication Lab	0	0	3	1.5	3	Practical	SDC (Minor)
86	ece	Drone Electronics	Minor	NECM401	UAV Sensors	3	1	0	4	4	Theory	DC (Minor)
87	ece	Drone Electronics	Minor	NECM402	Drone Electronics Lab	0	0	3	1.5	3	Practical	SDC (Minor)
88	ece	Drone Electronics	Minor	NECC306	Control Systems	3	1	0	4	4	Theory	DC (Minor)
89	ece	Drone Electronics	Minor	NECM403	Embedded Systems	3	1	0	4	4	Theory	DC (Minor)
90	ece	Drone Electronics	Minor	NECM404	Embedded System Lab	0	0	3	1.5	3	Practical	SDC (Minor)
91	ece	Electronics Engineering	Minor	NECC202	Signals & Networks	3	1	0	4	4	Theory	DC (Minor)
92	ece	Electronics Engineering	Minor	NECM303	Signals & Electronic Networks Lab	0	0	3	1.5	3	Practical	SDC (Minor)
93	ece	Electronics Engineering	Minor	NECC206	Analog Circuits	3	1	0	4	4	Theory	DC (Minor)
94	ece	Electronics Engineering	Minor	NECM304	Electronic Circuits Lab	0	0	3	1.5	3	Practical	SDC (Minor)
95	ece	Electronics Engineering	Minor	NECC203	Microprocessors & Microcontrollers	3	1	0	4	4	Theory	DC (Minor)
96	ece	Electronics Engineering	Minor	NECM301	Microprocessor Lab	0	0	3	1.5	3	Practical	SDC (Minor)
97	ece	Electronics Engineering	Minor	NECC306	Control Systems	3	1	0	4	4	Theory	DC (Minor)
98	ece	Electronics Engineering	Minor	NECC305	Digital Signal Processing	3	1	0	4	4	Theory	DC (Minor)
99	ece	Electronics Engineering	Minor	NECM405	Signal Processing and Simulation Lab	0	0	3	1.5	3	Practical	SDC (Minor)
100	ee	Electrical Engineering	Honor	NEEH301	Microprocessors and Microcontrollers	3	1	0	4	4	Theory	DC (Hons)
101	ee	Electrical Engineering	Honor	NEEH302	Microprocessors and Microcontrollers Laboratory	0	0	3	1.5	3	Practical	DP (Hons)
102	ee	Electrical Engineering	Honor	NEEH303	Electrical Measurements	3	1	0	4	4	Theory	DC (Hons)
103	ee	Electrical Engineering	Honor	NEEH304	Electrical Measurements Laboratory	0	0	3	1.5	3	Practical	DP (Hons)
104	ee	Electrical Engineering	Honor	NEEH401	Modern Control	3	1	0	4	4	Theory	DC (Hons)
105	ee	Electrical Engineering	Honor	NEEH402	Modern Control Laboratory	0	0	3	1.5	3	Practical	DP (Hons)
106	ee	Electrical Engineering	Honor	NEEH403	Special Purpose Electric Machines and Drives	3	1	0	4	4	Theory	DC (Hons)
107	ee	Electrical Engineering	Honor	NEEH404	Reactive Power and Voltage Control	3	1	0	4	4	Theory	DC (Hons)
108	ee	Electrical Engineering	Honor	NEEH405	Power and Switchgear Laboratory	0	0	3	1.5	3	Practical	DP (Hons)

SL	dept_id	course_title	minor/honor	sub_code	sub_name	lecture	tutorial	practical	credits	contact_hours	sub_type	Course Type
109	ee	Electrical Technology	Minor	NEEC201	Signals, Systems and Networks	3	1	0	4	4	Theory	DC (Minor)
110	ee	Electrical Technology	Minor	NEEM301	Computer Aided Simulation of Electrical Systems Laboratory	0	0	3	1.5	3	Practical	SDC (Minor)
111	ee	Electrical Technology	Minor	NEEC206	Electrical Machines I	3	1	0	4	4	Theory	DC (Minor)
112	ee	Electrical Technology	Minor	NEEM302	Embedded Systems and PCB Design Laboratory	0	0	3	1.5	3	Practical	SDC (Minor)
113	ee	Electrical Technology	Minor	NEEC302	Power Electronics	3	1	0	4	4	Theory	DC (Minor)
114	ee	Electrical Technology	Minor	NEEM401	Virtual Instrumentation using GUI based Programming Laboratory	0	0	3	1.5	3	Practical	SDC (Minor)
115	ee	Electrical Technology	Minor	NEEC207	Control System Engineering	3	1	0	4	4	Theory	DC (Minor)
116	ee	Electrical Technology	Minor	NEEC208	Power System Engineering	3	1	0	4	4	Theory	DC (Minor)
117	ee	Electrical Technology	Minor	NEEM402	Software based Electromagnetic Design Laboratory	0	0	3	1.5	3	Practical	SDC (Minor)
118	ese	Environmental Engineering	Honor	NESH301	Industrial Wastewater Engineering	3	1	0	4	4	Theory	DC (Hons)
119	ese	Environmental Engineering	Honor	NESH302	Industrial Wastewater Engineering Practical	0	0	3	1.5	3	Practical	DP (Hons)
120	ese	Environmental Engineering	Honor	NESH303	Aerosols in the Atmosphere	3	1	0	4	4	Theory	DC (Hons)
121	ese	Environmental Engineering	Honor	NESH304	Aerosols in the Atmosphere Practical	0	0	3	1.5	3	Practical	DP (Hons)
122	ese	Environmental Engineering	Honor	NESH401	Indoor Air Pollution	3	1	0	4	4	Theory	DC (Hons)
123	ese	Environmental Engineering	Honor	NESH402	Indoor Air Pollution Practical	0	0	3	1.5	3	Practical	DP (Hons)
124	ese	Environmental Engineering	Honor	NESH403	Atmospheric Science and Climate Change	3	1	0	4	4	Theory	DC (Hons)
125	ese	Environmental Engineering	Honor	NESH404	Environmental Nanotechnology	3	1	0	4	4	Theory	DC (Hons)
126	ese	Environmental Engineering	Honor	NESH405	Environmental Nanotechnology Practical	0	0	3	1.5	3	Practical	DP (Hons)
127	ese	Environmental Engineering	Minor	NESC203	Introduction to Ecology and Environmental Microbiology	3	1	0	4	4	Theory	DC (Minor)
128	ese	Environmental Engineering	Minor	NESM301	Applied Environmental Microbiology Practical	0	0	3	1.5	3	Practical	SDC (Minor)
129	ese	Environmental Engineering	Minor	NESM302	Water and Air Pollution	3	1	0	4	4	Theory	DC (Minor)
130	ese	Environmental Engineering	Minor	NESM303	Water and Air Pollution Practical	0	0	3	1.5	3	Practical	SDC (Minor)
131	ese	Environmental Engineering	Minor	NESC301	Water Resource Engineering	3	1	0	4	4	Theory	DC (Minor)
132	ese	Environmental Engineering	Minor	NESM401	Applied Water Resource Engineering Practical	0	0	3	1.5	3	Practical	SDC (Minor)
133	ese	Environmental Engineering	Minor	NESC305	Solid Waste Management	3	1	0	4	4	Theory	DC (Minor)
134	ese	Environmental Engineering	Minor	NESC306	Environmental Impact Assessment	3	1	0	4	4	Theory	DC (Minor)
135	ese	Environmental Engineering	Minor	NESM402	Applied Solid Waste Management Practical	0	0	3	1.5	3	Practical	SDC (Minor)
136	fme	Mineral and Metallurgical Engineering	Honor	NFMH301	Characterisation of Materials	3	1	0	4	4	Theory	DC (Hons)
137	fme	Mineral and Metallurgical Engineering	Honor	NFMH302	Characterisation of Materials Laboratory	0	0	3	1.5	3	Practical	DP (Hons)
138	fme	Mineral and Metallurgical Engineering	Honor	NFMH303	Hydrometallurgy for Critical Metal Extraction	3	1	0	4	4	Theory	DC (Hons)
139	fme	Mineral and Metallurgical Engineering	Honor	NFMH304	Hydrometallurgy Laboratory	0	0	3	1.5	3	Practical	DP (Hons)
140	fme	Mineral and Metallurgical Engineering	Honor	NFMH401	Materials Handling Systems	3	1	0	4	4	Theory	DC (Hons)
141	fme	Mineral and Metallurgical Engineering	Honor	NFMH402	Materials Handling Systems Laboratory	0	0	3	1.5	3	Practical	DP (Hons)
142	fme	Mineral and Metallurgical Engineering	Honor	NFMH403	Modelling in Mineral Engineering	3	1	0	4	4	Theory	DC (Hons)
143	fme	Mineral and Metallurgical Engineering	Honor	NFMH404	Metal Forming and Processing Technology	3	1	0	4	4	Theory	DC (Hons)
144	fme	Mineral and Metallurgical Engineering	Honor	NFMH405	Simulation in Mineral Engineering Laboratory	0	0	3	1.5	3	Practical	DP (Hons)
145	fme	Metallurgy	Minor	NFMM301	Extractive and Process Metallurgy	3	1	0	4	4	Theory	DC (Minor)
146	fme	Metallurgy	Minor	NFMM302	Process Metallurgy Laboratory	0	0	3	1.5	3	Practical	SDC (Minor)
147	fme	Metallurgy	Minor	NFMC306	Mechanical Metallurgy	3	1	0	4	4	Theory	DC (Minor)
148	fme	Metallurgy	Minor	NFMM303	Mechanical Behaviour of Materials Laboratory	0	0	3	1.5	3	Practical	SDC (Minor)
149	fme	Metallurgy	Minor	NFMM401	Physical Metallurgy	3	1	0	4	4	Theory	DC (Minor)
150	fme	Metallurgy	Minor	NFMM402	Physical Metallurgy Laboratory	0	0	3	1.5	3	Practical	SDC (Minor)
151	fme	Metallurgy	Minor	NFMM403	Sustainable Metallurgy	3	1	0	4	4	Theory	DC (Minor)
152	fme	Metallurgy	Minor	NFMM404	Processing of Materials	3	1	0	4	4	Theory	DC (Minor)
153	fme	Metallurgy	Minor	NFMM405	Design and Processing of Materials Laboratory	0	0	3	1.5	3	Practical	SDC (Minor)
154	mc	Mathematics and Computing	Honor	NMCH301	Computer Networks	3	1	0	4	4	Theory	DC (Hons)
155	mc	Mathematics and Computing	Honor	NMCH302	Computer Networks Laboratory	0	0	3	1.5	3	Practical	DP (Hons)
156	mc	Mathematics and Computing	Honor	NMCH303	Software Engineering	3	1	0	4	4	Theory	DC (Hons)
157	mc	Mathematics and Computing	Honor	NMCH304	Software Engineering Laboratory	0	0	3	1.5	3	Practical	DP (Hons)
158	mc	Mathematics and Computing	Honor	NMCH401	Mathematical Modeling for Control Systems	3	1	0	4	4	Theory	DC (Hons)
159	mc	Mathematics and Computing	Honor	NMCH402	Mathematical Modeling for Control Systems Laboratory	0	0	3	1.5	3	Practical	DP (Hons)
160	mc	Mathematics and Computing	Honor	NMCH403	Introduction to Functional Analysis	3	1	0	4	4	Theory	DC (Hons)
161	mc	Mathematics and Computing	Honor	NMCH404	Multivariate Statistical Methods	3	1	0	4	4	Theory	DC (Hons)
162	mc	Mathematics and Computing	Honor	NMCH405	Multivariate Statistical Methods Laboratory	0	0	3	1.5	3	Practical	DP (Hons)

SL	dept_id	course_title	minor/honor	sub_code	sub_name	lecture	tutorial	practical	credits	contact_hours	sub_type	Course Type
163	mc	Mathematics and Computing	Minor	NMCC201	Algebra	3	1	0	4	4	Theory	DC (Minor)
164	mc	Mathematics and Computing	Minor	NMCM301	Computing Laboratory 1	0	0	3	1.5	3	Practical	SDC (Minor)
165	mc	Mathematics and Computing	Minor	NMCC206	Database Management Systems	3	1	0	4	4	Theory	DC (Minor)
166	mc	Mathematics and Computing	Minor	NMCM302	Computing Laboratory - DBMS	0	0	3	1.5	3	Practical	SDC (Minor)
167	mc	Mathematics and Computing	Minor	NMCC302	Numerical Methods	3	1	0	4	4	Theory	DC (Minor)
168	mc	Mathematics and Computing	Minor	NMCM401	Computing Laboratory - NM	0	0	3	1.5	3	Practical	SDC (Minor)
169	mc	Mathematics and Computing	Minor	NMCC207	Real Analysis	3	1	0	4	4	Theory	DC (Minor)
170	mc	Mathematics and Computing	Minor	NMCM402	Statistical Modeling and Simulation	3	1	0	4	4	Theory	DC (Minor)
171	mc	Mathematics and Computing	Minor	NMCM403	Statistical Modeling and Simulation Laboratory	0	0	3	1.5	3	Practical	SDC (Minor)
172	me	Mining Engineering	Honor	NMNH301	Mine Environmental Engineering	3	1	0	4	4	Theory	DC (Hons)
173	me	Mining Engineering	Honor	NMNH302	Mine Environmental Engineering Practical	0	0	3	1.5	3	Practical	DP (Hons)
174	me	Mining Engineering	Honor	NMNH303	Seabed and Extra-Terrestrial Mining	3	1	0	4	4	Theory	DC (Hons)
175	me	Mining Engineering	Honor	NMNH304	Rock Mechanics Practical -II	0	0	3	1.5	3	Practical	DP (Hons)
176	me	Mining Engineering	Honor	NMNH401	Geospatial Technology in Mining	3	1	0	4	4	Theory	DC (Hons)
177	me	Mining Engineering	Honor	NMNH402	Geospatial Technology Practical	0	0	3	1.5	3	Practical	DP (Hons)
178	me	Mining Engineering	Honor	NMNH403	Numerical Modelling for Mining Application	3	1	0	4	4	Theory	DC (Hons)
179	me	Mining Engineering	Honor	NMNH404	Coalbed Methane	3	1	0	4	4	Theory	DC (Hons)
180	me	Mining Engineering	Honor	NMNH405	Coalbed Methane Practical	0	0	3	1.5	3	Practical	DP (Hons)
181	me	Mining Engineering	Minor	NMNC203	Surface Mining	3	1	0	4	4	Theory	DC (Minor)
182	me	Mining Engineering	Minor	NMNM301	Basic Mining Technology Practical	0	0	3	1.5	3	Practical	SDC (Minor)
183	me	Mining Engineering	Minor	NMNC208	Underground Coal Mining	3	1	0	4	4	Theory	DC (Minor)
184	me	Mining Engineering	Minor	NMNM302	Basic Mine Model Practical	0	0	3	1.5	3	Practical	SDC (Minor)
185	me	Mining Engineering	Minor	NMNC301	Underground Metal Mining	3	1	0	4	4	Theory	DC (Minor)
186	me	Mining Engineering	Minor	NMNM401	Basic Mine Surveying Practical	0	0	3	1.5	3	Practical	SDC (Minor)
187	me	Mining Engineering	Minor	NMNC207	Mine Ventilation	3	1	0	4	4	Theory	DC (Minor)
188	me	Mining Engineering	Minor	NMNC306	Mine Legislation and Safety	3	1	0	4	4	Theory	DC (Minor)
189	me	Mining Engineering	Minor	NMNM402	Basic Mine Ventilation Practical	0	0	3	1.5	3	Practical	SDC (Minor)
190	mech	Mechanical Engineering	Honor	NMEH301	Advanced Solid Mechanics	3	1	0	4	4	Theory	DC (Hons)
191	mech	Mechanical Engineering	Honor	NMEH302	Advanced Design Lab	0	0	3	1.5	3	Practical	DP (Hons)
192	mech	Mechanical Engineering	Honor	NMEH303	Modeling of Dynamical Systems	3	1	0	4	4	Theory	DC (Hons)
193	mech	Mechanical Engineering	Honor	NMEH304	Modeling of Dynamical Systems Lab	0	0	3	1.5	3	Practical	DP (Hons)
194	mech	Mechanical Engineering	Honor	NMEH401	NC and Computer Integrated Manufacturing	3	1	0	4	4	Theory	DC (Hons)
195	mech	Mechanical Engineering	Honor	NMEH402	Computer Integrated Manufacturing Lab	0	0	3	1.5	3	Practical	DP (Hons)
196	mech	Mechanical Engineering	Honor	NMEH403	Smart Manufacturing	3	1	0	4	4	Theory	DC (Hons)
197	mech	Mechanical Engineering	Honor	NMEH404	Fundamentals of Turbulence in Shear Flows	3	1	0	4	4	Theory	DC (Hons)
198	mech	Mechanical Engineering	Honor	NMEH405	Aerodynamics And Aeroacoustics Lab	0	0	3	1.5	3	Practical	DP (Hons)
199	mech	Mining Machinery Engineering	Honor	NMMH301	Hybrid Hydraulic Transmission Systems	3	1	0	4	4	Theory	DC (Hons)
200	mech	Mining Machinery Engineering	Honor	NMMH302	Hydraulic Systems Lab	0	0	3	1.5	3	Practical	DP (Hons)
201	mech	Mining Machinery Engineering	Honor	NMMH303	Condition Monitoring of Machines	3	1	0	4	4	Theory	DC (Hons)
202	mech	Mining Machinery Engineering	Honor	NMMH304	Condition Monitoring of Machines Lab	0	0	3	1.5	3	Practical	DP (Hons)
203	mech	Mining Machinery Engineering	Honor	NMMH401	Mine Automation	3	1	0	4	4	Theory	DC (Hons)
204	mech	Mining Machinery Engineering	Honor	NMMH402	Mine Automation Lab	0	0	3	1.5	3	Practical	DP (Hons)
205	mech	Mining Machinery Engineering	Honor	NMMH403	Design of Mine Equipment Components	3	1	0	4	4	Theory	DC (Hons)
206	mech	Mining Machinery Engineering	Honor	NMMH404	Advanced Material Handling Systems	3	1	0	4	4	Theory	DC (Hons)
207	mech	Mining Machinery Engineering	Honor	NMMH405	Material Handling Lab	0	0	3	1.5	3	Practical	DP (Hons)
208	mech	Mechanical Engineering	Minor	NMEM301	Mechanical Engineering I	3	1	0	4	4	Theory	DC (Minor)
209	mech	Mechanical Engineering	Minor	NMEM302	Mechanical Engineering I Lab	0	0	3	1.5	3	Practical	SDC (Minor)
210	mech	Mechanical Engineering	Minor	NMEM303	Mechanical Engineering II	3	1	0	4	4	Theory	DC (Minor)
211	mech	Mechanical Engineering	Minor	NMEM304	Mechanical Engineering II Lab	0	0	3	1.5	3	Practical	SDC (Minor)
212	mech	Mechanical Engineering	Minor	NMEM401	Industrial Automation	3	1	0	4	4	Theory	DC (Minor)
213	mech	Mechanical Engineering	Minor	NMEM402	Industrial Automation Lab	0	0	3	1.5	3	Practical	SDC (Minor)
214	mech	Mechanical Engineering	Minor	NMEM403	Maintenance Engineering	3	1	0	4	4	Theory	DC (Minor)
215	mech	Mechanical Engineering	Minor	NMEM404	Fundamentals and Applications of Renewable Energy	3	1	0	4	4	Theory	DC (Minor)
216	mech	Mechanical Engineering	Minor	NMEM405	Renewable Energy Lab	0	0	3	1.5	3	Practical	SDC (Minor)

SL	dept_id	course_title	minor/honor	sub_code	sub_name	lecture	tutorial	practical	credits	contact_hours	sub_type	Course Type
217	mech	Mining Machinery Engineering	Minor	NMMC301	Underground Mining Equipment	3	1	0	4	4	Theory	DC (Minor)
218	mech	Mining Machinery Engineering	Minor	NMMM301	Mining Equipment Lab - I	0	0	3	1.5	3	Practical	SDC (Minor)
219	mech	Mining Machinery Engineering	Minor	NMMC206	Opencast Mining Equipment	3	1	0	4	4	Theory	DC (Minor)
220	mech	Mining Machinery Engineering	Minor	NMMM302	Mining Equipment Lab - II	0	0	3	1.5	3	Practical	SDC (Minor)
221	mech	Mining Machinery Engineering	Minor	NMMC302	Design of Machine Elements	3	1	0	4	4	Theory	DC (Minor)
222	mech	Mining Machinery Engineering	Minor	NMMM401	Mining Equipment Lab - III	0	0	3	1.5	3	Practical	SDC (Minor)
223	mech	Mining Machinery Engineering	Minor	NMMC207	Hydraulics and Pneumatics	3	1	0	4	4	Theory	DC (Minor)
224	mech	Mining Machinery Engineering	Minor	NMMC306	Mineral Processing and Handling Equipment	3	1	0	4	4	Theory	DC (Minor)
225	mech	Mining Machinery Engineering	Minor	NMMM402	Mining Equipment Lab - IV	0	0	3	1.5	3	Practical	SDC (Minor)
226	msie	Management	Minor	NMSM301	Fundamentals of Decision Modelling	3	1	0	4	4	Theory	DC (Minor)
227	msie	Management	Minor	NMSM302	Stochastic Systems Laboratory	0	0	3	1.5	3	Practical	SDC (Minor)
228	msie	Management	Minor	NMSM303	Fundamentals of Operations Management	3	1	0	4	4	Theory	DC (Minor)
229	msie	Management	Minor	NMSM304	Software Laboratory using LINGO	0	0	3	1.5	3	Practical	SDC (Minor)
230	msie	Management	Minor	NMSM401	Macroeconomic Theory and Policy	3	1	0	4	4	Theory	DC (Minor)
231	msie	Management	Minor	NMSM402	Application of Spreadsheet Modelling	0	0	3	1.5	3	Practical	SDC (Minor)
232	msie	Management	Minor	NMSM403	Fundamentals of Financial Management	3	1	0	4	4	Theory	DC (Minor)
233	msie	Management	Minor	NMSM404	Organizational Behavior and Management for Managers	3	1	0	4	4	Theory	DC (Minor)
234	msie	Management	Minor	NMSM405	Systems Modelling and Simulation Laboratory	0	0	3	1.5	3	Practical	SDC (Minor)
235	pe	Petroleum Engineering	Honor	NPEH301	Drilling System Design	3	1	0	4	4	Theory	DC (Hons)
236	pe	Petroleum Engineering	Honor	NPEH302	Advanced Petroleum Equipment Design	0	0	3	1.5	3	Practical	DP (Hons)
237	pe	Petroleum Engineering	Honor	NPEH303	Gas Engineering	3	1	0	4	4	Theory	DC (Hons)
238	pe	Petroleum Engineering	Honor	NPEH304	Advanced Well Test Simulation	0	0	3	1.5	3	Practical	DP (Hons)
239	pe	Petroleum Engineering	Honor	NPEH401	Reservoir Geomechanics	3	1	0	4	4	Theory	DC (Hons)
240	pe	Petroleum Engineering	Honor	NPEH402	Advanced EOR Simulation	0	0	3	1.5	3	Practical	DP (Hons)
241	pe	Petroleum Engineering	Honor	NPEH403	Petroleum Engineering Design	3	1	0	4	4	Theory	DC (Hons)
242	pe	Petroleum Engineering	Honor	NPEH404	Pipeline Engineering	3	1	0	4	4	Theory	DC (Hons)
243	pe	Petroleum Engineering	Honor	NPEH405	Advanced Unconventional Reservoir Simulation	0	0	3	1.5	3	Practical	DP (Hons)
244	pe	Petroleum Engineering	Minor	NPEC201	Drilling Technology - I	3	1	0	4	4	Theory	DC (Minor)
245	pe	Petroleum Engineering	Minor	NPEM301	Petroleum Drilling Laboratory	0	0	3	1.5	3	Practical	SDC (Minor)
246	pe	Petroleum Engineering	Minor	NPEC207	Applied Reservoir Engineering	3	1	0	4	4	Theory	DC (Minor)
247	pe	Petroleum Engineering	Minor	NPEM302	Core Analysis Laboratory	0	0	3	1.5	3	Practical	SDC (Minor)
248	pe	Petroleum Engineering	Minor	NPEC203	Petroleum Formation Evaluation	3	1	0	4	4	Theory	DC (Minor)
249	pe	Petroleum Engineering	Minor	NPEM401	Petroleum Product Characterization Laboratory	0	0	3	1.5	3	Practical	SDC (Minor)
250	pe	Petroleum Engineering	Minor	NPEC208	Production Engineering - I	3	1	0	4	4	Theory	DC (Minor)
251	pe	Petroleum Engineering	Minor	NPEC306	Enhanced Oil Recovery Techniques	3	1	0	4	4	Theory	DC (Minor)
252	pe	Petroleum Engineering	Minor	NPEM402	EOR Laboratory	0	0	3	1.5	3	Practical	SDC (Minor)
253	phy	Engineering Physics	Honor	NPHH301	Smart Materials and Devices	3	1	0	4	4	Theory	DC (Hons)
254	phy	Engineering Physics	Honor	NPHH302	Materials Physics Laboratory	0	0	3	1.5	3	Practical	DP (Hons)
255	phy	Engineering Physics	Honor	NPHH303	Quantum Computation and Information	3	1	0	4	4	Theory	DC (Hons)
256	phy	Engineering Physics	Honor	NPHH304	Quantum Simulation Laboratory	0	0	3	1.5	3	Practical	DP (Hons)
257	phy	Engineering Physics	Honor	NPHH401	Spectroscopic Techniques for Industries	3	1	0	4	4	Theory	DC (Hons)
258	phy	Engineering Physics	Honor	NPHH402	Advanced Spectroscopy Laboratory	0	0	3	1.5	3	Practical	DP (Hons)
259	phy	Engineering Physics	Honor	NPHH403	Condensed Matter Physics	3	1	0	4	4	Theory	DC (Hons)
260	phy	Engineering Physics	Honor	NPHH404	Lithography and Nano-device Fabrication	3	1	0	4	4	Theory	DC (Hons)
261	phy	Engineering Physics	Honor	NPHH405	2D/3D Materials Fabrication Laboratory	0	0	3	1.5	3	Practical	DP (Hons)
262	phy	Quantum Technology	Minor	NPHM301	Introduction to Quantum Computation and Information	3	1	0	4	4	Theory	DC (Minor)
263	phy	Quantum Technology	Minor	NPHM302	Quantum Computation Laboratory	0	0	3	1.5	3	Practical	SDC (Minor)
264	phy	Quantum Technology	Minor	NPHM303	Survey of Quantum Technologies and Applications	3	1	0	4	4	Theory	DC (Minor)
265	phy	Quantum Technology	Minor	NPHM304	Basic Programming Laboratory	0	0	3	1.5	3	Practical	SDC (Minor)
266	phy	Quantum Technology	Minor	NPHM401	Foundations of Quantum Technologies	3	1	0	4	4	Theory	DC (Minor)
267	phy	Quantum Technology	Minor	NPHM402	Basic Laboratory Course for Quantum Technologies	0	0	3	1.5	3	Practical	SDC (Minor)
268	phy	Quantum Technology	Minor	NPHM403	Introduction to Quantum Materials	3	1	0	4	4	Theory	DC (Minor)
269	phy	Quantum Technology	Minor	NPHM404	Introduction to Quantum Optics	3	1	0	4	4	Theory	DC (Minor)
270	phy	Quantum Technology	Minor	NPHM405	Quantum Materials Laboratory	0	0	3	1.5	3	Practical	SDC (Minor)